

Question ID: 8736334b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question
Data set A: 72,73,73,76,76
Data set B: 61,64,74,85,x

Data set A and data set B each contain 5 numbers. If the mean of data set A is equal to the mean of data set B, what is the value of x ?

- Answer**
- A. 77
 - B. 85
 - C. 86
 - D. 95

Correct Answer: C

Rationale
Choice C is correct. The mean of a data set is found by dividing the sum of the values in the data set by the number of values in the data set. Therefore, the mean of data set A is $\frac{72+73+73+76+76}{5}$, which simplifies to 74. The mean of data set B is represented by the equation $\frac{61+64+74+85+x}{5}$, or $\frac{284+x}{5}$. It's given that the mean of data set A is equal to the mean of data set B. Therefore, the equation $74 = \frac{284+x}{5}$ can be used to solve for x. Multiplying both sides of this equation by 5 yields $370 = 284 + x$. Subtracting 284 from both sides of this equation yields $86 = x$.

Choices A, B, and D are incorrect and may result from calculation errors.